

Research Statement – Sam Guessoum

Research Vision

My research centers on developing real-time, data-driven machine learning and optimization systems for complex, high-dimensional environments. I am specially interested in how sensing, modeling, and decision-support frameworks can be integrated to improve infrastructure resilience, sustainability, and resource efficiency particularly in least developed countries (LDCs), low-income countries (LICs), and the Global South. My work bridges theoretical optimization with applied machine learning, focusing on robustness, interpretability, and deployment under real-world constraints for example:

Graduate Qualifying Project – Industrial Sensor Optimization

In collaboration with Findability Sciences, I developed a data-driven optimization framework for sugarcane milling processes. Using Gaussian Process modeling and multi-objective optimization, our system balanced sugar yield, purity, and energy consumption under real industrial constraints. We reduced over 150 sensor variables into interpretable control features and achieved approximately 4.8% improvement in milling efficiency. This work strengthened my interest in combining predictive modeling with optimization to improve sustainability in large-scale industrial systems.

Master's Thesis – Combinatorial Optimization (Bin Packing Problem)

My thesis focuses on solving the NP-hard Bin Packing Problem using multi-objective and adaptive optimization strategies. I explore hybrid approaches combining heuristic methods, metaheuristics, and exact algorithms to balance computational efficiency and solution quality. This work examines how optimization under constraints can reduce waste, improve resource allocation, and support scalable logistics and infrastructure systems.

Brain-Computer Interface Research – Real-Time Neural Hotspot Detection

In collaboration with Precision Neuroscience, I worked on a real-time neural signal analysis system using high-density micro-ECoG data streams. The project focused on detecting functionally tuned cortical regions ('hotspots') by distinguishing sustained neural activity from transient noise. I developed feature extraction and signal aggregation pipelines and built a decision-support interface designed for high-pressure surgical environments. This work deepened my interest in robust real-time inference and interpretable ML systems.

Explainable AI for Multimodal EEG Analysis

I am currently completing a research proposal exploring hybrid CNN-Vision Transformer architectures for multimodal EEG analysis, integrating SHAP and Grad-CAM explainability techniques. The goal is to improve both predictive accuracy and clinical interpretability, addressing the trust and deployment gap in medical AI systems. This research reflects my broader interest in building transparent, reliable models for complex signal environments.

Sustainability and Infrastructure-Focused Projects

I developed SolarSight, a data-driven framework designed to support solar energy deployment in under-resourced regions. The system evaluates infrastructure readiness, affordability constraints, and accessibility factors to guide scalable renewable energy strategies which are basically data-driven economic modeling initiatives connected to World Bank programs, balancing equity and efficiency in infrastructure systems in under developed African countries. Additionally, I contributed to a smart mobility optimization project analyzing transportation accessibility and sustainability constraints in urban environments in collaboration with urban suburban transport company of Algiers including ETUSA, SNTF and SETRAM

Future Research Directions

Moving forward, I aim to develop machine learning and optimization frameworks that operate under uncertainty and streaming constraints while remaining interpretable and scalable. I am particularly interested in integrating real-time sensing with robust optimization for energy systems, infrastructure planning, and sustainable resource allocation. My long-term goal is to build systems that not only achieve strong predictive performance but also meaningfully support decision-making in complex societal environments.